

DEVOTOPIA — RAG-READY KNOWLEDGE BASE

Purpose: compact factual reference for answering user questions about the Devotopia graduation project. Use the terminology below. Do not invent missing facts.

1. What Devotopia is

Devotopia is an AI-driven platform for personalised learning paths and evidence-based technical hiring. Its core idea is to make learning and verification the same activity: a learner chooses a target role, receives a dependency-ordered roadmap, takes adaptive assessments, and earns verified skills that can be used for hiring.

Problem: learners struggle with sequencing, knowing when a topic is sufficiently learned, and producing trustworthy evidence of ability. Employers struggle with CVs containing mostly self-declared skills and low-signal screening. Devotopia connects both sides through verified skill records.

Core loop: target role → generated roadmap → prerequisite-based modules → adaptive assessment → verified skill on pass / remedial module on failure → job matching and skill-gap closure.

2. Main capabilities

Area	Important capabilities
Learning	Role-based curriculum generation; prerequisite ordering; adaptive assessment; remedial module injection; module unlocking; track certification.
Talent	PDF résumé upload/parsing; AI enhancement while preserving the original; verified skill passport; shareable public profile.
Hiring	Employer job postings; verified-skill matching; gap analysis; one-action gap closing into the roadmap; candidate ranking by verified performance.
Support	Grounded AI study assistant; semantic retrieval; sandboxed coding challenges; mock interview; generated audio summaries.
Platform	Arabic/English with RTL; light/dark themes; installable PWA; subscriptions; administration/moderation.

3. Adaptive assessment — highest-value facts

Each assessment has **5 multiple-choice questions**. The first question batch starts at medium difficulty. After each answer, the next difficulty is determined by the previous two answers: both correct → raise one level; both incorrect → lower one level; one correct/one incorrect → hold. Difficulty saturates at hard/easy rather than wrapping.

Weights: easy = 1.0, medium = 1.5, hard = 2.0. Weighted score = earned difficulty weight ÷ total available difficulty weight × 100. The pass threshold is **70%**.

Pass: module completed, dependent modules unlocked through the prerequisite graph, verified skill recorded; a track certificate is issued when every module is complete. **Fail:** missed questions/topics update running per-topic failure statistics.

Remedial loop: when the running fail percentage for a topic reaches **30%** and the current attempt fails, a shorter remedial module is generated from the specific missed topics and inserted before the failed module. Generation is queued; an in-memory queue is used if Redis is unavailable.

Important limitation: five questions is a small sample, and weights 1.0/1.5/2.0 are reasoned rather than psychometrically calibrated. Scores are comparable within a module, not across the platform.